

Quality Assurance & Testing Report

Manual Testing Documentation

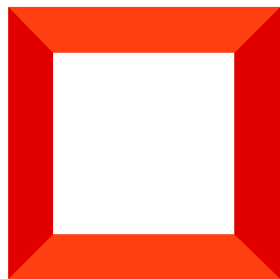
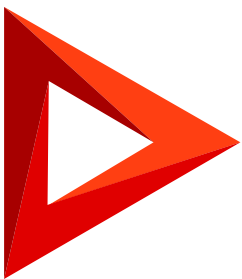


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Quality Assurance & Testing Report

1. Testing Methodologies

- **Functional Testing:** Verified all buttons, filters, modals, and endpoints perform their intended actions.
- **Security Testing (RBAC):** Validated JWT authentication and role-based controllers restrict unauthorized users (viewer role) from destructive actions.
- **Integration Testing:** Tested Node.js backend, MSSQL Server, and external APIs (VirusTotal, AbusIPDB) communication.
- **Performance & Real-Time Testing:** Simulated attacks via Inject Test Vector to verify Socket.io pushes updates in under 500ms.
- **UI/UX & Mobile Responsiveness:** Tested across Desktop, Tablet, and Mobile viewport sizes.

2. Complete Test Case Results

Module 1: Authentication & RBAC

ID	Module	Scenario	Status
TC-01	Auth	Login with valid Admin credentials redirects to Dashboard, JWT stored.	✓ Pass
TC-02	Auth	Login with invalid credentials displays error message.	✓ Pass
TC-03	RBAC	Access /api/audit as Viewer returns 403 Forbidden.	✓ Pass
TC-04	RBAC	Access Incident Page as Viewer page loads, New Incident button hidden.	✓ Pass
TC-05	Auth	Session expiry / Invalid Token auto-redirects to Login page.	✓ Pass

Module 2: Dashboard & Real-Time Sync

ID	Module	Scenario	Status
TC-06	Dash	Verify KPI metrics Critical, High, Open Incidents show correct numbers.	✓ Pass
TC-07	Socket	Inject Test Vector new alert appears without page refresh via Socket.io.	
TC-08	Dash	Threat Map correctly plots latitude/longitude from IP enrichment.	

Module 3: Alerts, Threat Intel, SOAR, Reports, Audit, Mobile

ID	Module	Scenario	Status
TC-09	Alerts	Filter by Critical severity table shows only Critical alerts.	✓ Pass
TC-10	Alerts	Date range filter displays only alerts within specified range.	
TC-11	Alerts	Acknowledge alert status changes to Acknowledged, logged in Audit.	
TC-12	Alerts	Escalate alert to incident modal opens, maps alert details correctly.	✓ Pass
TC-13	Threat	Auto IP enrichment backend fetches VirusTotal, AbuseIPDB, Geo data.	✓ Pass
TC-14	Threat	Manual IP scan displays VirusTotal gauge, malicious count, verdict.	
TC-15	Threat	Cache validation same IP within 1 hour returns fast SQL cached response.	

TC-16	Threat	Global blacklist combines AbuseIPDB data + local SOAR blocks.	✓ Pass
TC-17	SOAR	Block IP from Alert Detail appears in BlockedEntities and blacklist.	✓ Pass
TC-18	Reports	Download PDF no blank pages, valid tables and charts.	✓ Pass
TC-19	Reports	Download CSV matches PDF data, emojis format correctly.	✓ Pass
TC-20	Audit	Block IP action BLOCK_ENTITY, User ID, Timestamp logged immutably.	✓ Pass
TC-21	Audit	Audit log pagination Next/Previous work across large log sets.	✓ Pass
TC-22	UI	Mobile sidebar overlays content (z-index: 60), does not overlap Topbar.	✓ Pass
TC-23	UI	Alerts filter on mobile inputs stack vertically at full width, no overflow.	✓ Pass
TC-24	UI	Threat Intel form on mobile elements wrap correctly, no height collapse.	✓ Pass

3.Security & Compliance

3.1 Security Controls

Control		Implementation Detail
SQL Injection Prevention		All database queries use mssql parameterized inputs (@param). Raw string concatenation is never used in query building.
JWT Authentication		Stateless token-based authentication. Tokens are verified on every protected request via authMiddleware.js. Expired tokens auto-redirect to login.
Role-Based Control	Access	Roles: Admin, SOC Manager, Analyst, Viewer. The authorize() middleware restricts endpoints by role. Viewer cannot delete, create, or view audit logs.

Audit Trail	Every mutation (Login, Alert Acknowledge, IP Block, Incident changes) creates a permanent record in AuditLogs with User ID, Action, IP, and timestamp.
Threat Intel Rate Limiting	VirusTotal and AbuseIPDB responses are cached in ThreatFeeds SQL table. Duplicate scan requests within the cache window return stored data.

3.2 Role Permissions Matrix

Action / Feature	Admin	SOC Manager	Analyst	Viewer
View Dashboard & Alerts	✓	✓	✓	✓
Create / Update Incidents	✓	✓	✓	✗
Delete Incidents	✓	✓	✗	✗
Block IP / Domain (SOAR)	✓	✓	✓	✗
Unblock IP / Domain	✓	✓	✗	✗
View Audit Trail	✓	✓	✗	✗
Manage Users (/api/users)	✓	✓	✗	✗
Generate Reports	✓	✓	✓	✗